

Case Study: Sugar Bliss Bakery

Amit owns a newly opened bakery named **Sugar Bliss**, which sells different varieties of chocolates. To attract customers during festive seasons, the bakery offers **discounts on selected chocolate flavours**. Amit wants to automate the billing process by developing a **C# console-based application** that calculates the final payable amount after applying the appropriate discount.

You are required to help Amit by developing this application using **C# classes and objects**.

Requirements

Class Specification

Create a class named **Chocolate** with the following **public properties**:

Data Type	Property Name
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string	Flavour
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int	Quantity
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int	PricePerUnit
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double	TotalPrice
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double	DiscountedPrice
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Method in Chocolate Class (5 Marks)

```
public bool ValidateChocolateFlavour()
```

- This method validates the chocolate flavour.
 - If the flavour is **Dark**, **Milk**, or **White**, return `true`.
 - Otherwise, return `false`.
 - **Note:** Flavour is case-sensitive.
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Method in Program Class (10 Marks)

```
public Chocolate CalculateDiscountedPrice(Chocolate chocolate)
```

This method should:

- Calculate the total price
- Apply the discount based on the selected flavour
- Return the updated Chocolate object

Discount Details

Flavour Discount Percentage

Dark 18%

Milk 12%

White 6%

Price Calculation Formula

- **Total Price** = Quantity × PricePerUnit
 - **Discounted Price** = Total Price – (Total Price × Discount Percentage / 100)
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Main Method Instructions

Accept user input for:

- Chocolate flavour
 - Quantity
 - Price per unit
2. Call `ValidateChocolateFlavour()`:
 - If the method returns `false`, display **“Invalid flavour”**
 3. If the flavour is valid:
 - Call `CalculateDiscountedPrice()`
 - Display the following details:
 - Flavour
 - Quantity
 - Price per unit
 - Total price
 - Discounted price
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Notes

- All classes, properties, and methods must be **public**
 - Do **not** use `Environment.Exit()`
 - Do **not** modify method names or return types
 - Follow proper **object-oriented programming principles**
 - Output must match the sample format exactly
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Sample Input & Output

Sample Input 1

```
Enter the flavour
Dark
Enter the quantity
5
Enter the price per unit
20
```

Sample Output 1

```
Flavour : Dark
Quantity : 5
Price Per Unit : 20
Total Price : 100
Discounted Price : 82
```

Sample Input 2

```
Enter the flavour
Fruit
Enter the quantity
8
Enter the price per unit
12
```

Sample Output 2

```
Invalid flavour
```